

EXPERIMENTS

1. Pulses and their spectra
2. Nyquist: zero ISI
3. Live eye diagram
4. The matched filter
5. Matched and mismatched chains
6. BER: simulation meets theory
7. NRZ versus PAM-4
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CONTROLS

TRANSMITTER AND RECEIVER

Signal

2-PAM (NRZ)

4-PAM

Filters

RRC → RRC (matched)

Roll-off β

0.35

IMPAIRMENTS

Es/N0

25.0 dB

Timing jitter (rms)

0.0 % UI

☐ Band-limited channel

Channel bandwidth

0.5 × Rs

☒ Persistence

Tip: double-click a slider to reset it; type into a value to set it.

Live eye diagram — An oscilloscope in persistence mode: watch noise, jitter and bandwidth close the eye.

Eye height

89 %

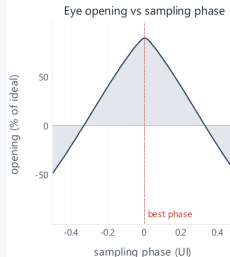
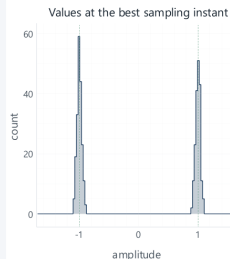
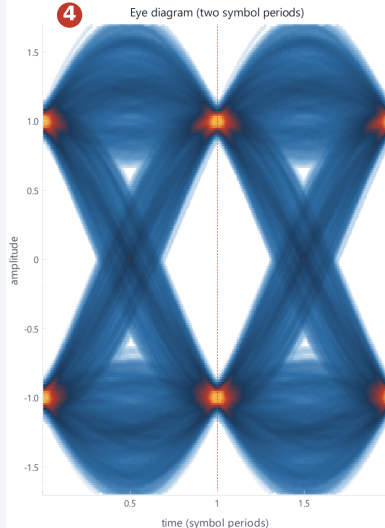
Eye width

66 % UI

SNR at the sampler

28.1 dB

Estimated BER

< 10^{-12} 

WHAT'S GOING ON

Reading an eye

Each trace is two symbol periods of the received waveform, all drawn on top of each other like a scope in persistence mode (brighter = more traces). Right now the eye is **wide open**: 89 % of its ideal height and 66 % of a symbol wide.

Vertical opening is your noise margin; **horizontal opening** is your timing margin. The receiver samples at the widest point (red line, bottom right).

Press Play/Pause to freeze the scope. In SerDes compliance testing, a hexagonal **mask** must fit inside the eye.

TRY THIS

0 / 4 done

- ☐ Squeeze the channel to 0.25 Rs or less and still keep the eye at least 50 % open.
- ☐ Close the eye horizontally with jitter alone: width below 50 % at Es/N0 ≥ 30 dB, no channel.
- ☐ 4-PAM: keep all three eyes at least 50 % open with Es/N0 of 19 dB or less.
- ☐ See pure ISI: choose RRC → nothing with Es/N0 ≥ 35 dB. The traces no longer meet at single points.

Read more in the book

Chapter 8, §8.8 Eye diagrams (p. 385)